POWER BI PRACTICAL ASSIGNMENT

Healthcare Data Cleaning & Star Schema Modelling

A complete step-by-step guide: Power Query cleaning, AgeGroup, dimension tables, fact table and relationships

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Dataset: healthcare_dataset(2).csv (55,500 rows, 15 columns)

Tool: Power BI Desktop — Power Query Editor and Model View

Output: A cleaned star-schema data model ready for clinical analytics

OVERVIEW

What You Will Build

Follow the 27 parts in order. Do not skip a part — later steps depend on earlier ones.

Parts	What you do
1–3	Load the CSV and create the base table **Healthcare_Cleaned**
4–9	Set data types, clean text, standardize Name, remove duplicates, create **AgeGroup**, standardize **Admission Type**
10–14	Create the patient, doctor and hospital dimension tables with IDs
15–20	Create **FactAdmission** and replace descriptive columns with IDs
21–22	Create **DimDate** and review the final queries
23–27	Close & Apply, create relationships, final star schema and requirement check

PART 1

Load the CSV into Power BI

Step 1

Open **Power BI Desktop**.

Go to:

> **Home → Get Data → Text/CSV**

Select the file:

healthcare_dataset(2).csv

Then click **Transform Data**. This opens the **Power Query Editor**.

PART 2

Create the Main Cleaned Table

In the left side of Power Query you will see a query called:

healthcare_dataset

Rename it to:

Healthcare_Cleaned

Right-click the query → **Rename**. This will be our **base table**.

PART 3

Check the Columns

Your table should contain these 15 columns:

#	Column
1	Name
2	Age
3	Gender
4	Blood Type
5	Medical Condition
6	Date of Admission
7	Doctor
8	Hospital
9	Insurance Provider
10	Billing Amount
11	Room Number
12	Admission Type
13	Discharge Date
14	Medication
15	Test Results

PART 4

Set the Correct Data Types

This is important. Set each column to the data type shown below.

Column	Data Type
Name	Text
Age	Whole Number
Gender	Text
Blood Type	Text
Medical Condition	Text
Date of Admission	Date
Doctor	Text
Hospital	Text
Insurance Provider	Text
Billing Amount	Decimal Number
Room Number	Whole Number
Admission Type	Text
Discharge Date	Date
Medication	Text
Test Results	Text

For the dates

Select **Date of Admission**, then go to:

> **Transform → Data Type → Date**

Do the same for **Discharge Date**.

> **Tip:** If Power BI asks whether to replace the current type, choose **Replace Current**.

PART 5

Clean Text Columns

We want to remove unnecessary spaces. Select the text columns:

- Name
- Gender
- Blood Type
- Medical Condition
- Doctor
- Hospital
- Insurance Provider
- Admission Type
- Medication
- Test Results

Then apply, one after the other:

> **Transform → Format → Trim**

> **Transform → Format → Clean**

This removes unwanted spaces and non-printing characters.

PART 6

Standardize Name

The dataset contains names with inconsistent capitalization. For example:

Before (messy)	After (clean)
Bobby JacksOn	Bobby Jackson
LesLie TErRy	Leslie Terry
DaNnY sMitH	Danny Smith

Select the **Name** column, then go to:

> **Transform → Format → Capitalize Each Word**

This makes the names consistent and supports your data-cleaning requirement.

PART 7

Remove Exact Duplicate Records

The CSV has **55,500 rows**, of which **534 are exact duplicate rows**. These are the records we remove.

How

In Power Query, go to:

> **Home → Remove Rows → Remove Duplicates**

You do **not** need to select only Name. For this step Power Query checks the **entire row**. If two rows have exactly the same Name, Age, Gender, ... Test Results, one copy is removed.

> **Important:** Do **NOT** remove duplicates based only on Name. The dataset has repeated names, and a repeated name does not necessarily mean a duplicate admission.

PART 8

Create AgeGroup

Assignment requirement: *Group patient ages into logical demographic categories via Power Query.*

Go to:

> **Add Column → Conditional Column**

New column name: `AgeGroup`

Create these rules:

If Age is less than 18 → **Pediatric**

Else if Age is less than 40 → **Young Adult**

Else if Age is less than 60 → **Middle Adult**

Else → **Senior**

So your categories are:

Age	AgeGroup
0–17	Pediatric
18–39	Young Adult
40–59	Middle Adult
60+	Senior

Your table now contains, for example:

Name	Age	AgeGroup
Bobby Jackson	30	Young Adult
Leslie Terry	62	Senior
Danny Smith	76	Senior

PART 9

Standardize Admission Type

The dataset already has exactly three admission types: **Emergency**, **Elective** and **Urgent**. Still, we clean the field.

Select **Admission Type**, then apply:

> **Transform → Format → Trim**

> **Transform → Format → Capitalize Each Word**

Your final values should be:

- Emergency
- Elective
- Urgent

This satisfies the requirement: *Standardize admission types (Emergency, Elective, Urgent).*

PART 10

Now Create the Star Schema

We now have a cleaned base table: **Healthcare_Cleaned**. From it we will create five tables:

- DimPatient
- DimDoctor
- DimHospital
- DimDate
- FactAdmission

> **Important issue:** This dataset has **no Patient ID**, so we need to create one carefully (Parts 11–12).

PART 11

Create DimPatient

Right-click **Healthcare_Cleaned → Reference**. Rename the new query to:

DimPatient

You now have another query containing the same cleaned data.

Keep these columns

- Name
- Age
- AgeGroup
- Gender
- Blood Type

Select these columns → right-click → **Remove Other Columns**.

PART 12

Create PatientID

Because the source has no Patient ID, we need a **surrogate ID**. Do **not** simply add an Index to DimPatient yet — the **same PatientID must also appear in FactAdmission**.

The safest student-project approach is to build the patient table from the unique patient attributes. For this dataset, the patient-identifying combination is:

- Name
- Age
- Gender
- Blood Type

1. Select these four columns: **Name, Age, Gender, Blood Type**.
2. Go to **Home → Remove Rows → Remove Duplicates** Each unique patient profile now appears once.
3. Go to **Add Column → Index Column → From 1**
4. Rename **Index** → **PatientID**.

Your table becomes:

PatientID	Name	Age	Gender	Blood Type
1	Bobby Jackson	30	Male	B-
2	Leslie Terry	62	Male	A+
3	Danny Smith	76	Female	A-

Keep **AgeGroup** too. The final DimPatient columns are:

PatientID · Name · Age · AgeGroup · Gender · Blood Type

PART 13

Create DimDoctor

Go back to **Healthcare_Cleaned**. **Do not** create this from DimPatient.

1. Right-click **Healthcare_Cleaned → Reference**.
2. Rename it **DimDoctor**.
3. Keep only the column **Doctor** (Remove Other Columns).
4. **Home → Remove Rows → Remove Duplicates**
5. **Add Column → Index Column → From 1**
6. Rename **Index** → **DoctorID**.

Final DimDoctor columns:

DoctorID · Doctor

PART 14

Create DimHospital

Again, start from **Healthcare_Cleaned**.

1. Right-click **Healthcare_Cleaned → Reference**.
2. Rename it **DimHospital**.
3. Keep only the column **Hospital** (Remove Other Columns).
4. **Home → Remove Rows → Remove Duplicates**
5. **Add Column → Index Column → From 1**
6. Rename **Index → HospitalID**.

Final DimHospital columns:

HospitalID · Hospital

PART 15

Create FactAdmission

Right-click **Healthcare_Cleaned → Reference** once more and rename the new query to:

FactAdmission

Keep the admission-related columns (Remove Other Columns):

- Name
- Age
- Gender
- Blood Type
- Doctor
- Hospital
- Date of Admission
- Discharge Date
- Admission Type
- Medical Condition
- Insurance Provider
- Billing Amount
- Room Number
- Medication
- Test Results

We will then replace the descriptive fields with IDs where appropriate.

PART 16

Create AdmissionID

In **FactAdmission**, go to:

> **Add Column → Index Column → From 1**

Rename **Index** → **AdmissionID**. Every admission record now has a unique number:

AdmissionID	Name	Admission Type	Billing
1	Bobby Jackson	Urgent	18856
2	Leslie Terry	Emergency	33643
3	Danny Smith	Emergency	27955

> **Remember:** This ID represents the **admission record**, not the patient.

PART 17

Add PatientID to FactAdmission

This is the important part. Currently:

FactAdmission has Name, Age, Gender, Blood Type.
DimPatient has PatientID, Name, Age, AgeGroup, Gender, Blood Type.

We need to match the records and bring **PatientID** into FactAdmission.

1. Go to **Home → Merge Queries**
2. **First table:** FactAdmission
3. **Second table:** DimPatient
4. Select the matching columns — because the dataset has no source Patient ID, use **Name, Age, Gender, Blood Type**. Select them **in the same order in both tables**.
5. **Join Kind:** Left Outer. Click **OK**.
6. A new **DimPatient** column appears. Click its **expand** button.
7. Select only **PatientID** and click **OK**.

FactAdmission now contains **PatientID**.

PART 18

Add DoctorID

1. In FactAdmission: **Home → Merge Queries**
2. **Second table:** DimDoctor
3. Match **FactAdmission[Doctor]** with **DimDoctor[Doctor]**.
4. **Join Kind:** Left Outer. Click **OK**.
5. Expand the new column and select **DoctorID**.

FactAdmission now has **DoctorID**.

PART 19

Add HospitalID

1. In FactAdmission: **Home → Merge Queries**
2. **Second table:** DimHospital
3. Match **FactAdmission[Hospital]** with **DimHospital[Hospital]**.
4. **Join Kind:** Left Outer. Click **OK**.
5. Expand the new column and select **HospitalID**.

FactAdmission now has **HospitalID**.

PART 20

Remove Unnecessary Descriptive Columns from FactAdmission

Once the IDs are added, you no longer need these repeated descriptive fields in the fact table. **Remove** them:

- Name
- Age
- Gender
- Blood Type
- Doctor
- Hospital

Keep these 13 columns:

- AdmissionID
- PatientID
- DoctorID
- HospitalID
- Date of Admission
- Discharge Date
- Admission Type
- Medical Condition
- Insurance Provider
- Billing Amount
- Room Number
- Medication
- Test Results

Your FactAdmission should now look like this (conceptually):

AdmissionID	PatientID	DoctorID	HospitalID	Admission Type	Billing
1	1	25	7	Urgent	18856
2	2	18	15	Emergency	33643
3	3	43	21	Emergency	27955

PART 21

Create DimDate

This one is easier.

1. In Power Query: **Home → New Source → Blank Query**
2. Rename the query to **DimDate**.
3. Go to **Home → Advanced Editor**
4. Delete everything in the editor and paste the code below, then click **Done**.

```
let
    StartDate = List.Min(Healthcare_Cleaned[Date of Admission]),
    EndDate = List.Max(Healthcare_Cleaned[Date of Admission]),
    DateList = List.Dates(
        StartDate,
        Duration.Days(EndDate - StartDate) + 1,
        #duration(1,0,0,0)
    ),
    DateTable = Table.FromList(
        DateList,
        Splitter.SplitByNothing(),
        {"Date"}
    ),
    ChangedType = Table.TransformColumnTypes(
        DateTable,
        {{"Date", type date}}
    ),
    AddYear = Table.AddColumn(
        ChangedType,
        "Year",
        each Date.Year([Date]),
        Int64.Type
    ),
    AddMonth = Table.AddColumn(
        AddYear,
        "Month",
        each Date.MonthName([Date]),
        type text
    ),
    AddMonthNumber = Table.AddColumn(
        AddMonth,
        "MonthNumber",
        each Date.Month([Date]),
        Int64.Type
    ),
    AddQuarter = Table.AddColumn(
        AddMonthNumber,
        "Quarter",
        each "Q" & Text.From(Date.QuarterOfYear([Date])),
        type text
    )
in
    AddQuarter
```

This creates DimDate with the columns:

Date · Year · Month · MonthNumber · Quarter

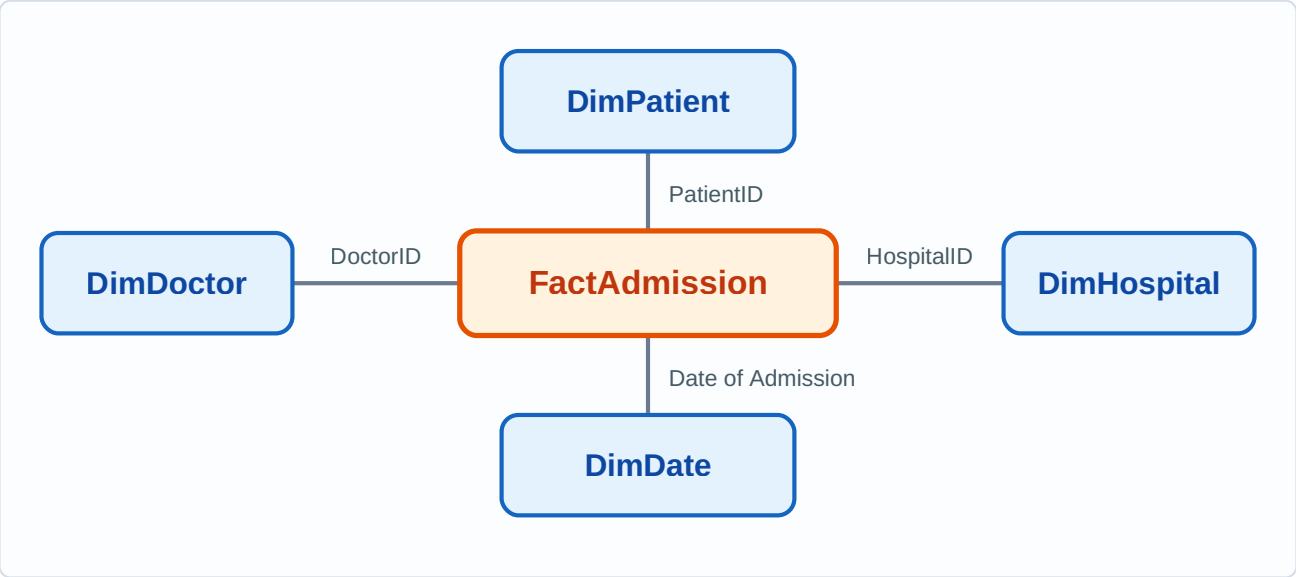
PART 22

Final Power Query Tables

At this point, your left panel should contain these six queries:

- Healthcare_Cleaned
- DimPatient
- DimDoctor
- DimHospital
- FactAdmission
- DimDate

The structure is:



PART 23

Close & Apply

Once all the transformations are complete, go to:

> **Home → Close & Apply**

Power BI will load the tables into the data model.

PART 24

Create Relationships

Go to **Model View**. You should see the tables. Create these four relationships:

Relationship	'One' side (dimension)	'Many' side (fact)
Patient	DimPatient[PatientID]	FactAdmission[PatientID]
Doctor	DimDoctor[DoctorID]	FactAdmission[DoctorID]
Hospital	DimHospital[HospitalID]	FactAdmission[HospitalID]
Date	DimDate[Date]	FactAdmission[Date of Admission]

For **all four** relationships:

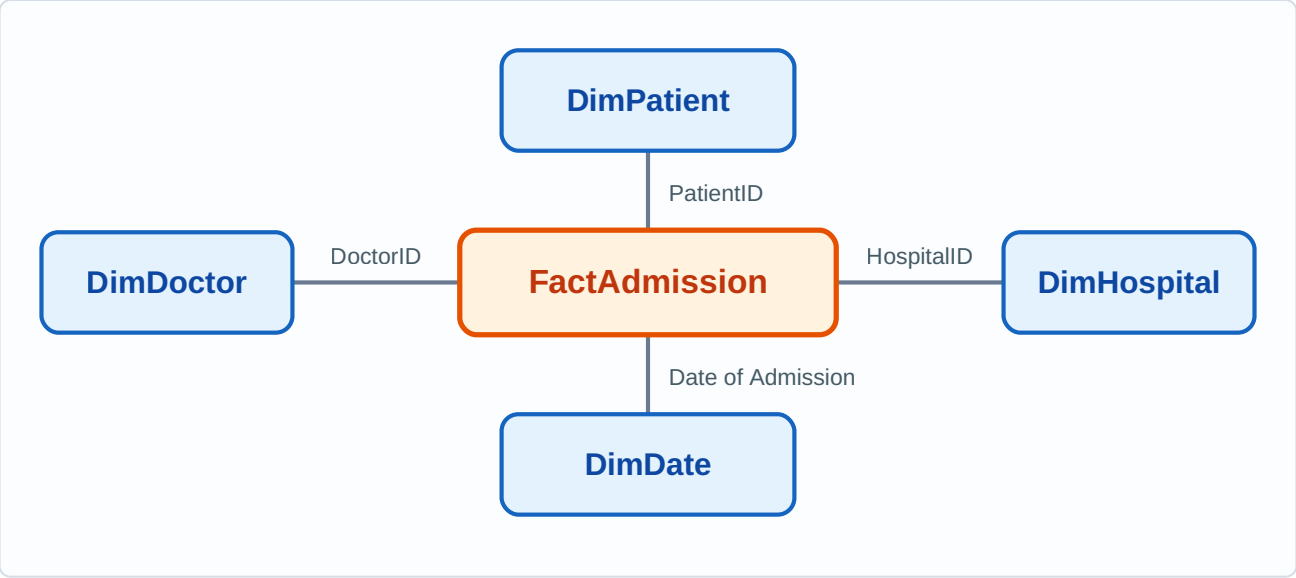
- **Cardinality** = One to many (1:*)
- **Cross filter direction** = Single

> **Tip:** Each relationship has the **1** on the dimension table and the * on FactAdmission. If Power BI auto-detects any relationship the wrong way round, double-click the line and correct it.

PART 25

Your Final Star Schema

It should look approximately like this:



FactAdmission sits in the centre. Each dimension connects to it through its ID (DimDate connects through the date).

PART 26

What Each Table Means

This is the easiest way to remember it:

Table	Purpose
DimPatient	**Who** is the patient?
DimDoctor	**Who** is the doctor?
DimHospital	**Where** did the patient receive care?
DimDate	**When** did the admission happen?
FactAdmission	**What** admission happened, and what were its measurable details?

PART 27

What Your Assignment Has Now Achieved

Assignment requirement	Done through
Group patient ages into logical demographic categories via Power Query	✓ AgeGroup created (Part 8)
Standardize admission types (Emergency, Elective, Urgent)	✓ Admission Type cleaned and standardized (Part 9)
Set up star-schema model relationships between patient dimensions and admission facts	✓ DimPatient → FactAdmission ✓ DimDoctor → FactAdmission ✓ DimHospital → FactAdmission ✓ DimDate → FactAdmission (Part 24)
Output a cleaned data model ready for clinical analytics	✓ Cleaned data ✓ Duplicate records removed ✓ Correct data types ✓ Patient dimension ✓ Doctor dimension ✓ Hospital dimension ✓ Date dimension ✓ Admission fact table ✓ Relationships

CHECKLIST

Before You Submit

- ☐ Query renamed to Healthcare_Cleaned; data types set correctly (Part 4)
- ☐ Text columns trimmed and cleaned; Name capitalized (Parts 5–6)
- ☐ 534 exact duplicate rows removed using the whole row (Part 7)
- ☐ AgeGroup column created with 4 categories (Part 8)
- ☐ Admission Type standardized (Part 9)
- ☐ DimPatient, DimDoctor, DimHospital each have an ID starting from 1 (Parts 11–14)
- ☐ FactAdmission has AdmissionID, PatientID, DoctorID, HospitalID (Parts 16–19)
- ☐ Descriptive columns removed from FactAdmission (Part 20)
- ☐ DimDate created with Advanced Editor code (Part 21)
- ☐ Close & Apply done; four 1:* single-direction relationships in Model View (Parts 23–24)